

TW02: Environment Setup and Jupyter

Five challenges | 20 marks each | 100 marks total

Student name	
Student ID	
Class / group	
Submission date	

Weekly purpose

Use Anaconda or Python safely, understand paths and packages, work reliably in Jupyter, and diagnose beginner errors.

SUBMISSION RULES

- Complete work in the corresponding Jupyter notebook.
- Write your own code and explanations; no solutions are supplied.
- Keep supplied data unchanged and work on copies.
- Restart the kernel and run all cells before submission.
- Submit the notebook with outputs visible and assistance acknowledged.

MARKING RUBRIC FOR EACH CHALLENGE

Assessment area	Marks
Correct method and working code	8
Validation and quality checks	4
Explanation and interpretation	4
Reproducibility and presentation	4
Total per challenge	20

Challenge 1: Create an environment identity card

Scenario

A tutor needs enough information to diagnose a student's Python setup without seeing the computer.

INSTRUCTIONS

1. Display the Python version without printing the complete build string.
2. Display the active Python executable, operating system, and current working folder.
3. Check whether the current folder contains the supplied data directory.
4. Store the results in a dictionary called `environment_report`.
5. Explain which executable should be checked when Jupyter uses the wrong environment.

REQUIRED SUBMISSION

- A labelled `environment_report` dictionary.
- A Boolean data-folder check.
- A short troubleshooting explanation.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 2: Navigate files with pathlib

Scenario

A notebook must find a text file using a relative path so it works on another student's computer.

INSTRUCTIONS

1. Create a Path for `week2_environment_notes.txt` inside `DATA_DIR`.
2. Check that the path exists and is a file before reading it.
3. Display its name, suffix, parent folder, and size in bytes.
4. Read the UTF-8 text and count its lines.
5. Explain why an absolute path copied from your computer is not portable.

REQUIRED SUBMISSION

- A valid relative Path and four file properties.
- The complete text and correct line count.
- A portability explanation.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 3: Audit the required packages

Scenario

Before teaching starts, you must verify that NumPy, pandas, SQLAlchemy, and JupyterLab are available.

INSTRUCTIONS

1. Use `importlib.util.find_spec` to check each required package without importing it.
2. Use `importlib.metadata.version` to report installed versions when available.
3. Record package, installed status, and version in a small table or dictionary.
4. Write the conda or pip command needed for any missing package.
5. Explain why installation should target the same environment used by the notebook kernel.

REQUIRED SUBMISSION

- A four-package audit.
- A suitable installation command if anything is missing.
- A kernel/environment explanation.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 4: Repair four beginner errors

Scenario

A classmate's notebook contains a `NameError`, `TypeError`, `SyntaxError`, and a hidden-state problem.

INSTRUCTIONS

1. Read each broken example stored as text so it does not stop the notebook.
2. Copy each example into a new cell and correct it.
3. For every error, name the error category and explain the cause in one sentence.
4. Demonstrate one safe type conversion.
5. Restart the kernel and run all cells to confirm that hidden state has been removed.

REQUIRED SUBMISSION

- Four corrected code examples.
- Four error names and cause explanations.
- A statement confirming a successful restart-and-run-all test.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Challenge 5: Build a reproducible mini-notebook

Scenario

A short analysis must be understandable and repeatable by another beginner.

INSTRUCTIONS

1. Add a Markdown title, business question, method, result, and conclusion.
2. Calculate the mean and median of the supplied response times using the statistics module.
3. Round displayed results to two decimal places without changing the raw values.
4. Add at least two assertions that confirm the count and valid range.
5. Restart and run all cells, then record whether the notebook completed without errors.

REQUIRED SUBMISSION

- A professionally structured notebook section.
- Correct mean and median values.
- Two passing assertions and a reproducibility statement.

20 marks: method and working code 8 | validation and quality checks 4 | explanation and interpretation 4 | reproducibility and presentation 4

Final submission checklist

- ☐ All five challenges are complete.
- ☐ The notebook runs from top to bottom without errors.
- ☐ Every output is clearly labelled.
- ☐ Important results include written explanations.
- ☐ Validation checks are included.
- ☐ Supplied raw data has not been overwritten.
- ☐ Sources and assistance are acknowledged.

Student declaration

I confirm that this submission is my work, that I have acknowledged sources and assistance, and that the displayed outputs were produced by the submitted code.

Signature: _____ Date: _____